

H. KAAK KALE

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EDUCATION

Georgia Institute of Technology

Ph.D. in Machine Learning, School of Electrical and Computer Engineering

Research interests: Large Language Models, State Space Models,
Mechanistic Interpretability, Privacy, and Efficient Training

2026 – Present
Atlanta, GA, USA

Boğaziçi University

B.Sc. in Electrical & Electronics Engineering.
Specialization: Signal Processing

2020 - 2025
Istanbul, Turkey

Istanbul University State Conservatory

Part-Time Studies Majoring in Piano Performance.

2012 - 2021
Istanbul, Turkey

PUBLICATIONS

- Jana, P., Kale, K., Tanriverdi, A. E., Song, C., Vishwanath, S., & Ganesh, V. **ProofBridge: Auto-Formalization of Natural Language Proofs in Lean via Joint Embeddings**. In *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
- Baser, O., Kale, K., Li, P.-H., & Chinchali, S. **Fair Resource Allocation for Fleet Intelligence**. In *2025 IEEE Global Communications Conference (GLOBECOM)*, pp. 4952-4957. IEEE, 2025.
- Baser, O., Tanriverdi, A. E., Kale, K., Chinchali, S., & Vishwanath, S. **WavShape: Information-Theoretic Speech Representation Learning for Fair and Privacy-Aware Audio Processing**. In *Proceedings of Interspeech 2025*, pp. 4873-4877, 2025.
- Kale, K., Esfahanizadeh, H., Elias, N., Baser, O., Médard, M., & Vishwanath, S. **TexShape: Information Theoretic Sentence Embedding for Language Models..** In *2024 IEEE International Symposium on Information Theory (ISIT)*, pp. 2038-2043. IEEE, 2024.
- Baser, O., Kale, K., & Chinchali, S. (2024). **SecureSpectra: Safeguarding Digital Identity from Deep Fake Threats via Intelligent Signatures**. In *Proceedings of Interspeech 2024*, 1115-1119, doi: 10.21437/Interspeech.2024-2349.
- Akcin, O., Tanriverdi, A. E., Kale, K., & Chinchali, S. **Fleet Supervisor Allocation: A Submodular Maximization Approach**. In *8th Annual Conference on Robot Learning 2024 (CoRL)*.
- Baser, O., Kale, K., Li, P.-H., & Chinchali, S. **Fair Resource Allocation for Fleet Intelligence**. In *2025 IEEE Global Communications Conference (GLOBECOM)*, pp. 4952-4957. IEEE, 2025.
- Elias, N., Esfahanizadeh, H., Kale, K., Vishwanath, S., & Médard, M. **MultiTok: Variable-Length Tokenization for Efficient LLMs Adapted from LZW Compression**. Poster, *Learn to Compress & Compress to Learn Workshop, IEEE International Symposium on Information Theory (ISIT)*, 2026.

RESEARCH & WORK EXPERIENCE

Protopia AI

Research Engineer Intern

May 2026 – August 2026
London, UK

- Developed large-scale training and fine-tuning infrastructure for **NVIDIA Nemotron-3 Ultra**, implementing expert, context/sequence, tensor, and fully-sharded data parallelism for distributed training on multi-GPU systems.
- Profiled and optimized long-context training memory usage, identifying major bottlenecks in KNN-based metrics and unnecessary logit materialization; restructured metric computation and introduced mechanisms to bypass the language-model head when logits were not required.
- Implemented CPU activation offloading and additional memory optimizations, enabling substantially longer training sequences and supporting experiments beyond previous context-length limits.
- Investigated numerical instability in long-context **Mamba/Transformer** training, including NaN failures emerging at large sequence lengths, and developed instrumentation for monitoring Mamba states, training dynamics, and chunk-level utility metrics.
- Extended privacy-preserving LLM infrastructure through **TurboQuant**, output-protection and proxy integrations, early **VisionCloak** privacy evaluations, and shared-model / Jigsaw training infrastructure.
- Contributed production-ready distributed-training and model-infrastructure changes across Protopia's training and core ML repositories, including expert parallelism, LM-head sharding, memory-efficient utility metrics, and model-hook infrastructure.

Protopia AI
Research Contractor (Remote)

November 2024 – December 2025
Austin, TX, USA

- Researched privacy-preserving representations for large language models, developing and evaluating learned transformations that protect sensitive information while preserving downstream model utility.
- Developed **BeamClean**, a language-aware embedding reconstruction attack combining parameter estimation with language-model-guided decoding to evaluate the robustness of embedding obfuscation mechanisms.
- Contributed to the **Stained Glass Transform**, including privacy evaluation, reconstruction attacks, mutual-information-based metrics, and utility-privacy analysis for transformed LLM embeddings.
- Built scalable experimentation and evaluation infrastructure for privacy research, including reconstruction metrics, model inversion experiments, memory optimization, and large-scale PyTorch training workflows.

The University of Texas at Austin, LINC Lab
Privacy Aware Robust Learning in Networked Systems
Worked on-site from June 2024 - September 2024.
Supervisor: Prof. Sriram Vishwanath

August 2023 - December 2024
Austin, TX, USA

The University of Texas at Austin, Swarm Lab
Privacy Aware Robust Learning in Networked Systems
Worked on-site from June 2024 - September 2024.
Supervisor: Prof. Sandeep Chinchali

June 2023 - December 2024
Austin, TX, USA

PROJECTS

Designing a Passive Radar Using FM Radio for Target Detection
Developing a covert radar system using ambient FM/TV signals for aircraft tracking, enhanced by machine learning techniques for improved detection.
Supervisor: Prof. Ali Emre Pusane

September 2024 - Present

LLM Privacy/ Coded LLM
Developed a framework for achieving privacy-preserving and efficient large language model pipelines.
Supervisor: Prof. Sriram Vishwanath, Prof. Muriel Médard

August 2023 - August 2024

Earnings Call Speech Acoustic Analysis
Developing a framework to align speech recordings with imperfect transcripts
Analysis of specific acoustic features of speech elements.
Supervisor: Prof. Erdinc Akyildirim

November 2023 - Present

Ultrasonic Distance Measurement System
Developed a VHDL application for precise distance measurements using an ultrasonic sensor.

May 2023 - June 2023

HONORS & AWARDS

- **Merit and Need-Based Scholarship**
Institution: Tekfen Holding
Description: Received for outstanding academic performance and financial need.
September 2020 - Ongoing
- Ranked **301st** among 2.2 million students taking the Turkish university entrance exam. June 2020
- Secured **3rd** place nationally in the Fizy National Music Contest with the school band. April 2018
- Ranked **1st** in the Kemer Country Golf Club golf tournament. April 2018

SKILLS, ACTIVITIES & INTERESTS

- *Skills:*
Python, PyTorch, Large Language Models, Linux, Information Theory, Representation Learning, TensorFlow, Matlab, VHDL, Arduino, LaTeX, C++
 - *Interests:*
Privacy, Digital Signal Processing, Decentralized Computing, Computer Vision, Music, Piano.
- *Languages:*
Native in Turkish
Fluent in English (TOEFL iBT Score: **112/120**)
Intermediate in Italian
 - *Other Extracurriculars:*
Playing Piano & Guitar, Tennis, Golf

ADDITIONAL EDUCATION & INTERESTS

Istanbul University State Conservatory
Part-Time Piano Performance Studies

2012 – 2021
Istanbul, Turkey

Languages: Turkish (Native), English (Fluent; TOEFL iBT 112/120), Italian (Intermediate)
Interests: Piano, Music Production, Tennis, Kiteboarding